PROJECT PROPOSAL

FunType

Typing-Speed Trainer

Train the hands, the words follow.

Honest. Instant. Private.

Prepared by	Kaon — developer and product owner
Revision	4 · 14 October 2026
Status	Submitted for approval
Decision requested by	25 September 2026
Live build	kaonhew02.github.io/FunType
Repository	github.com/KaonHew02/FunType

THE ASK

Approval for a ten-week Phase 4 — freeze and test the measurement model, triple the drill content, add history and export, and build per-key coaching — with one part-time developer (about 50 hours) and RM 0 committed.

Document control

Document	FunType — Project Proposal
Revision	4, 14 October 2026
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Baseline	Repository `KaonHew02/FunType` at commit `8b795dc` (14 October 2026); application code as of `8b795dc`
Source of truth	`docs/PROPOSAL.md` in the repository. This Word document is generated from it, so the two never disagree

Revision history

Rev	Date	Summary
1	22 Sep 2026	First issue — scope, measurement model, architecture, design, roadmap and risks (`6465d3b`)
2	23 Sep 2026	Brought up to the 23 September tamper guards (`a14402c`), then rewritten as a fourteen-section formal proposal (`c742315`)
3	23 Sep 2026	Restructured into the eighteen-section house format with a cover, document control, contents and three diagrams; Word and PDF editions; personas, milestones, three-year cost, sign-off and glossary added. **Corrected:** FunType is already live on GitHub Pages, so publishing is no longer a proposed step
4	14 Oct 2026	Brought up to the guard fix (`8b795dc`): the on-screen guard now compares text only and stands down for browser extensions, which had frozen the last second of a test. Risk R13 and test TC-19 added; sizes, line counts and line ranges refreshed. The plan and its dates are unchanged

How to read this proposal

Sections 1–4 make the case: what FunType is, the problem it answers, the measurement model at its core, and how success is measured. Sections 5–11 are the detail a reviewer needs to trust it — users, scope, architecture, data, security, design and stack. Sections 12–16 are the proposal proper: the plan, how it is tested, what could go wrong, what it costs and what comes after. Section 17 asks for the decision, and section 18 is a glossary of the typing and technical terms used throughout.

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1 Executive summary

FunType is a browser-based typing-speed trainer: one HTML file, four drill modes, no server, no account and nothing to install. A working build is already live at kaonhew02.github.io/FunType — 929 lines of hand-written HTML, CSS and JavaScript in a single 46 KB file, with no build step and no libraries. As of 23 September 2026 it also ignores scripted keystrokes and pasted text, seals its saved records against hand-editing, and puts back any score edited on screen.

The product exists because the trainers people already use either score generously or want an account, and often both — and a typing score is only worth practising against if the typist can trust how it was worked out.

Product	FunType — *Train the hands, the words follow.*
Category	Typing-speed trainer (web application)
Primary user	People who type for a living — developers, writers, support staff — and students practising for speed
Delivered	4 drill modes, 2 difficulty toggles, a five-part measurement model, a pace chart, per-configuration personal bests, 8 themes
Stack	One static HTML file on GitHub Pages · vanilla JavaScript · the browser's `localStorage`
Security	Trusted-keystroke check · paste and drop blocked · sealed save · on-screen score guard · 400 WPM cap
Dependencies	No npm packages, no build step, no framework, no API keys; one Google Fonts stylesheet
Running cost	RM 0/month — hosting is GitHub Pages and storage is the user's own browser
Effort to date	6 commits, 11 Sep 2026 – 14 Oct 2026
Proposed next phase	10 weeks: freeze and test the measurement model, triple the content, add history, build per-key coaching

The ask. Approval to run the ten-week Phase 4 in Project plan and timeline (§12) at the resourcing set out in Resources and budget (§15) — one developer, part-time (about 50 hours), and RM 0 committed.

2 Background and problem statement

The problem is not that typing trainers do not exist — it is that most of them measure in a way that rewards the wrong habit, or put an account between the user and the first keystroke. A typing score is a promise about skill. If the arithmetic behind it is generous, the typist practises toward a number rather than toward accuracy.

FunType began on 11 September 2026 as a single-file drill page. The scope question arrived early: a speed figure is worthless without a definition, and a definition is worthless if the stored best was produced under a different one. The project therefore treats the measurement model — not the modes or the themes — as the product.

2.1 What the alternatives get wrong

Option	What it costs the user
Gamified school trainers	Gross WPM counts every keystroke, right or wrong, so the score climbs as the typist gets sloppier; badges and class accounts come first
Platform-style trainers	A login, leaderboards and several menus before the first keystroke; results kept on someone else's server
Trainers that charge for backspace	Fixing a mistake lowers the score, which teaches people to leave mistakes in
One "best" across every setting	Turning punctuation on drops WPM, so the record shows which setting was easiest, not how fast the person types
A stopwatch and a paragraph	Private and free, but no per-character accuracy, no pace curve and no record

2.2 The developer gap

Most trainers drill English prose. A working developer's speed ceiling is not on prose — it is on brackets, quotes, semicolons, `=>` and the shift key. A trainer that never shows `data.filter(r => r.active)` never trains the characters that actually slow a programmer down.

2.3 The design flaw underneath all of it

Most trainers keep one headline number and change what feeds it without saying so. When the definition shifts — gross to net, counting spaces or not, charging backspace or not — every stored best stops being comparable with every new run, and there is no way to recompute it because the keystrokes were never kept. FunType's founding constraint, *define the number in public and never change it silently*, exists to make that class of error impossible.

3 Proposed solution

FunType is one page that measures typing the way a sceptic would, and refuses everything else. There is no account to create, nothing to install and nothing sent anywhere.

3.1 From keystroke to personal best

Every keystroke passes one gate, changes one piece of state, and is scored from that state alone. Nothing downstream writes back upstream: the results screen and the records bar only ever display what scoring produced.

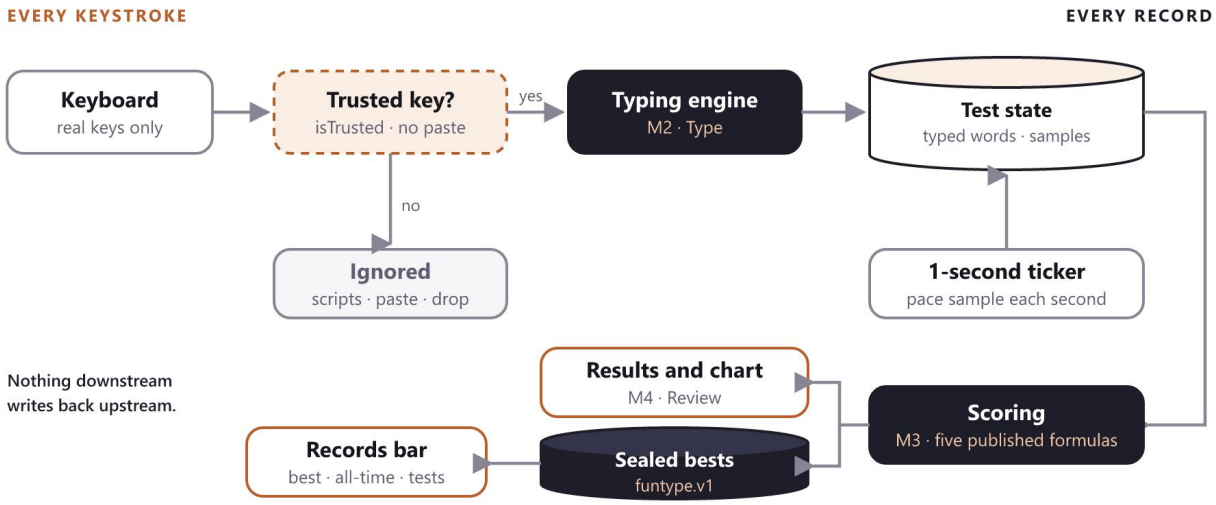


Figure 1 — From keystroke to personal best — one gate, one state, one scorer

Scripted keys stop at the gate. Pasted text never reaches the engine at all. A best is written only by scoring, only when it beats the stored figure, and only when it is humanly possible.

3.2 The measurement model

Five figures, each with one published formula. They are implemented exactly as stated here and in the repository README.

Net WPM — the headline. Only correct characters count. A correctly typed word also earns its trailing space; a word left wrong does not.

```
net wpm = correct characters ÷ 5 ÷ minutes
```

Raw WPM — shown beside it, on purpose. Every character typed, right or wrong, over the same clock. The gap between raw and net is the signature of typing faster than accuracy can support, which is the most common way to plateau.

```
raw wpm = all characters typed ÷ 5 ÷ minutes
```

Accuracy — backspaces are free. A backspace is neither a keystroke nor an error, so correcting a mistake never costs score.

```
accuracy = correct keystrokes ÷ (correct + wrong keystrokes) × 100
```

Consistency — how even the pace was. Each second the engine records characters typed that second × 12, a WPM-equivalent; consistency is how little that series varies, clamped to 0–100.

```
consistency = 100 × (1 − standard deviation ÷ mean)
```

The character breakdown — reported as `correct / wrong / extra / missed`:

Class	Means	What it tells the typist
correct	The right character in the right place	—
wrong	A wrong character in a position	Finger accuracy
extra	Typed past the end of a word	Not reading ahead
missed	Space pressed before the word was finished	Anticipating the word boundary

Splitting *extra* from *missed* matters because they are opposite faults with opposite fixes.

3.3 One record per exact configuration

A personal best is stored against the exact settings that produced it. The key joins mode, length and toggles — `time-30`, `words-25-p`, `time-60-p-n`, `code-sql` — so switching punctuation on starts a fresh record rather than polluting the old one. A best is claimed on net WPM; accuracy and the date are stored beside it for context.

3.4 What the user gets that they did not have

- **A number they can trust.** Net WPM as the headline, raw beside it, and every formula published.
- **An instant start.** The words are on screen when the page opens, and the first keystroke starts the clock. No login, no splash, no setup.
- **Drills for real work.** A `code` mode on real JavaScript, Python and SQL, where a developer's speed actually sits.
- **Privacy by construction.** Records stay in one browser on one machine. There is no account, no analytics and nothing to delete later.
- **A file you can copy.** One HTML file that runs from a web host, a USB drive or an email attachment, with no toolchain to rot.

4 Objectives and success criteria

The project succeeds if a typist can practise on FunType daily for a month and never have reason to doubt a score. Everything below is a test, not an aspiration.

#	Objective	Measure	Threshold	Status
O1	The headline counts only correct work	Wrong characters counted in net WPM	0	**Met**
O2	Correcting a mistake is never punished	Backspaces counted as keystrokes or errors	0	**Met**
O3	Practice starts instantly	Time from page load to a counted keystroke; screens before it	≤ 2 s; 0 screens	**Met**
O4	Records stay comparable	Personal bests shared between different settings	0	**Met**
O5	The app runs with nothing installed	npm packages, build steps, frameworks, API keys	0 of each	**Met**
O6	Nothing about the user leaves the device	Requests carrying typing data or identifiers	0	**Met**
O7	Runs anywhere, unchanged	Opens from `file://`, a USB drive and GitHub Pages	3 of 3	**Met**
O8	A script cannot type	Synthetic key or input events accepted	0	**Met (23 Sep 2026)**
O9	A hand-edited save cannot hold a record	Edited saves accepted on load	0	**Met (23 Sep 2026)**
O10	Impossible runs never become records	Personal bests above 400 WPM	0	**Met (23 Sep 2026)**
O11	Motion can be switched off	Animations running under reduced-motion settings	0	**Met**
O12	Every theme is readable	Themes checked for legibility and a correct mark	8 of 8	**4 of 8 — dusk, daylight, bubblegum, terminal**
O13	The scoring is proven, not trusted	Automated assertions per metric	≥ 1 per metric	**Not met — no automated tests yet**
O14	The content cannot be memorised quickly	Unique words in the drill pool	≥ 1,000	**Not met — 288 today**

4.1 What "done" means for Phase 4

O13 and O14 are the two open rows that matter, and they come first in Phase 4 because every later feature — history, coaching — is only as good as the numbers underneath it. O12 is a visual pass that closes in the release week. Phase 4 then adds one new objective: a results screen that names the typist's weakest keys and drills them.

4.2 Explicit non-objectives

These are refused on purpose, and each refusal has a reason that should survive a change of mind:

- **No accounts or cloud sync.** Both need a server, and both move the user's records off the device.

- **No global leaderboard.** A leaderboard is only as honest as the server that re-checks each run. FunType has no server, so any leaderboard it showed would be the easiest thing in the product to fake.
- **No lessons or curriculum.** Guided home-row teaching is a different product with a different user.
- **No analytics or telemetry.** The privacy claim is only true if there is nothing to switch off.
- **No gross WPM headline, and no charge for backspace.** Both are exactly the habits the product exists to correct.

5 Target users and personas

The target user is one adult who already types every day and wants to get measurably faster — on a physical keyboard, at a desk. Not a classroom, not a child learning the home row, and not someone looking for a game.

Persona	Situation	What they need	Lives in
Aiman, 27 — the developer	Writes TypeScript all day; about 70 WPM on prose, much slower on brackets, arrows and quotes	Drills on real code, and a raw-versus-net gap that shows where the errors are	`code`, `time 60` with punctuation
Mei Ling, 31 — the support agent	Answers chat tickets all shift; a typo in a reply costs more than a slow reply	Accuracy that does not punish correcting, and a calm screen	`words 50`, `text medium`
Hafiz, 20 — the student	Practises on a shared college computer and will not make an account there	An instant start, nothing to sign up for, records that stay in that browser	`time 30`, `time 60`
Priya, 35 — the writer	Drafts long-form; bursts on familiar words and stalls on new ones	A consistency score that shows the stalls, and long passages	`text long`, `time 120`

One person is often two of these at once, which is why they are one trainer and not four.

5.1 Jobs to be done

- *When I have ten minutes between tasks, I want to start typing immediately, so that I actually practise instead of setting up.*
- *When my score drops, I want to see whether it was speed or accuracy, so that I know what to fix.*
- *When I switch on punctuation, I want a separate record, so that a harder drill does not look like a bad day.*
- *When I write code for a living, I want to drill the symbols, so that my working speed improves and not just my prose speed.*
- *When I use a shared computer, I want no account, so that nothing about me is left on a server.*

5.2 Who this is not for

Schools needing class management, assignments and teacher reports; beginners who need guided home-row lessons; and anyone who wants to race strangers on a public leaderboard. The last is a real and common want, and Future roadmap (§16) states plainly what it would take to meet it honestly.

6 Scope — module breakdown

Seven modules live inside the one file, each named by the verb it serves. All seven are delivered. Phase 4 deepens M1 and M5 and adds an eighth, M8 Coach.

#	Verb	Module	Main code	Status
M1	Drill	Modes and content	`WORDS`, `TEXTS`, `CODE`, `buildWords`, `decorate`	**Delivered**
M2	Type	Typing engine	`typeChar`, `commitWord`, `backspace`, `moveCaret`	**Delivered**
M3	Measure	Scoring	`tally`, `tick`, `consistency`	**Delivered**
M4	Review	Results and pace chart	`showResults`, `chartSVG`	**Delivered**
M5	Remember	Records and storage	`load`, `save`, `renderRecords`	**Delivered**
M6	Guard	Score integrity	trusted-event checks, `sealOf`, `draw`	**Delivered**
M7	Style	Themes and brand	`THEMES`, `applyTheme`, the keycap mark	**Delivered**
M8	Coach	Weak-key analysis	—	Proposed for Phase 4

6.1 M1 · Drill — Modes and content

Mode	Lengths	Content	What it trains
time	15 / 30 / 60 / 120 s	Common-word list; 140 words at the start, 60 more whenever fewer than 30 remain	Sustained speed against a clock
words	10 / 25 / 50 / 100 words	Common-word list	Finishing a fixed target
text	short / medium / long	6 written passages — 3 short, 2 medium, 1 long — about typing technique	Rhythm across real sentences
code	js / py / sql	9 real lines of code, 3 per language	Symbols, brackets and shift-key reach

- The word list holds 298 entries — 288 unique, averaging 4.4 characters — weighted to short, frequent words, with a tail of typing vocabulary such as *keyboard*, *rhythm* and *accuracy*. The same word never appears twice in a row.
- `! punctuation` marks about 27.5% of words: commas, full stops, question and exclamation marks, semicolons, colons, quotes, brackets and `'s`. The first word and every word after a full stop, `?` or `!` is capitalised.
- `# numbers` replaces about 11% of words with a number; 40% of those run to four digits.
- Both toggles apply to `time` and `words`. Passages and code are typed exactly as written.

6.2 M2 · Type — Typing engine

- The clock starts on the first keystroke, not on page load. While a test runs, the settings bar fades out and a live counter shows time left (or words done) and current WPM.
- Each character is coloured as it is typed — correct, wrong, or extra — and a word left wrong is underlined once passed.

- `space` commits a word and is ignored on an empty one, so double spaces cannot be typed.
- `backspace` can step back into the previous word **only if that word was left wrong**, which matches a real editor and keeps accuracy meaningful. `ctrl` + `backspace` clears the whole word.
- Extra characters stop at 12 past the end of a word, so a stuck key cannot flood the test.
- Three lines are visible; the text scrolls so the current line stays second from the top.
- After 8 seconds without a key the words blur behind a focus prompt. The clock keeps running.
- On-screen and IME keyboards that report every key as "Unidentified" are read through a hidden input that mirrors what was typed.

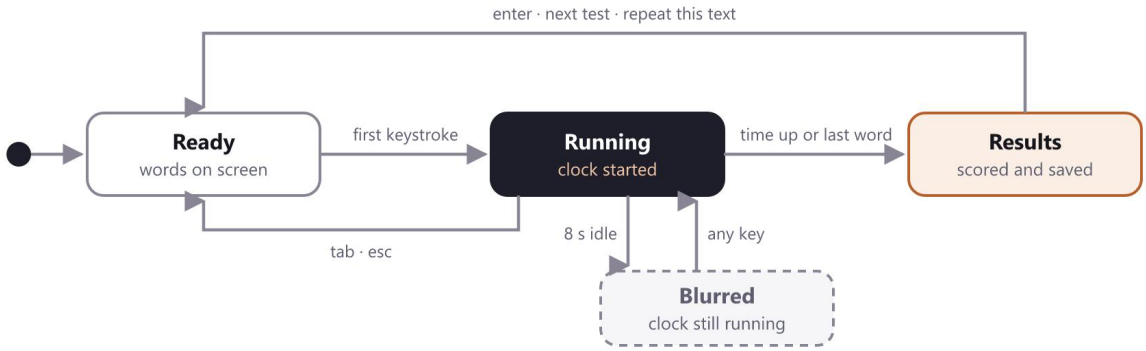


Figure 2 — The life of one test

`tab` restarts with the same words; `esc` deals new ones; `enter` on the results screen starts the next test. Clicking the logo restarts too — the brand mark is a working button.

6.3 M3 · Measure — Scoring

- A one-second ticker samples elapsed time, net and raw WPM, and the errors made in that second. The samples feed the live counter, the pace chart and the consistency score.
- The per-second pace step is characters typed that second × 12 — sixty seconds divided by five characters per word.
- Final figures are computed once, at the finish, from the whole test state, using the formulas in The measurement model (§3.2). Nothing is accumulated along the way that could drift from the record.
- Elapsed time has a floor of 0.1 s, so a test finished instantly cannot divide by zero.

6.4 M4 · Review — Results and pace chart

- **Headline:** net WPM and accuracy, with a badge — `new best · +N`, `first record`, or `not recorded · over 400 wpm`.
- **Pace chart:** net WPM (filled) and raw WPM per second, with a ✕ where errors landed. The axis snaps to a fixed ladder — 20, 25, 40, 50, 60, 80, 100, 120, 150, 200, 250, 300 — rather than to the data, so two runs can be compared by eye. A run with fewer than two samples shows a note instead of a chart.
- **Six facts:** the test settings, raw WPM, consistency, the character breakdown, time in seconds to one decimal, and the personal best.
- **Two buttons:** *next test* for fresh words, and *repeat this text* for the same words, to separate improvement from the luck of the draw.

6.5 M5 · Remember — Records and storage

- A best is stored per configuration key with its net WPM, accuracy and timestamp. A tie does not replace the stored best.
- The header carries three live chips: the best for the current setting, the all-time best across every setting, and the number of tests run.
- Mode, lengths, toggles and theme are remembered between visits.
- Every read and write is wrapped so blocked storage — a private window, a hardened browser — never stops the app. It runs normally and simply forgets afterwards.

The storage design is set out in Data model and storage design (§8).

6.6 M6 · Guard — Score integrity

Six guards against faked scores, delivered on 23 September 2026. They are listed with the threat each one answers in Security and privacy (§9).

6.7 M7 · Style — Themes and brand

Eight colour themes, the keycap mark and the wordmark. They are covered in Brand, UI and UX design (§10).

6.8 M8 · Coach — Weak-key analysis (proposed)

The engine already knows every keystroke that missed and how long each key took. M8 aggregates both across the last twenty runs into a per-character profile — *you lose most of your time on* `;`, `p` *and capitals* — and deals a drill weighted toward exactly those characters. It converts the measurement work from a scoreboard into coaching, and it is the main deliverable of Phase 4.

6.9 Out of scope

Excluded	Reason
Accounts, login and cloud sync	Need a backend, and move the user's records off the device
Global leaderboards and multiplayer races	Need a server that re-checks every run; without one they reward cheating
A mobile or touch-first layout	The product trains a physical keyboard; the layout stays usable on a phone but is not designed for one
Lessons, curriculum and teacher dashboards	A classroom product with a different user and a different shape
Advertising and analytics	Contradict the privacy claim

7 System architecture

FunType is a client-only application. One HTML file carries 144 lines of CSS, 66 lines of markup and 701 lines of JavaScript; the browser is the runtime and `localStorage` is the only database. GitHub Pages serves the file and does nothing else.

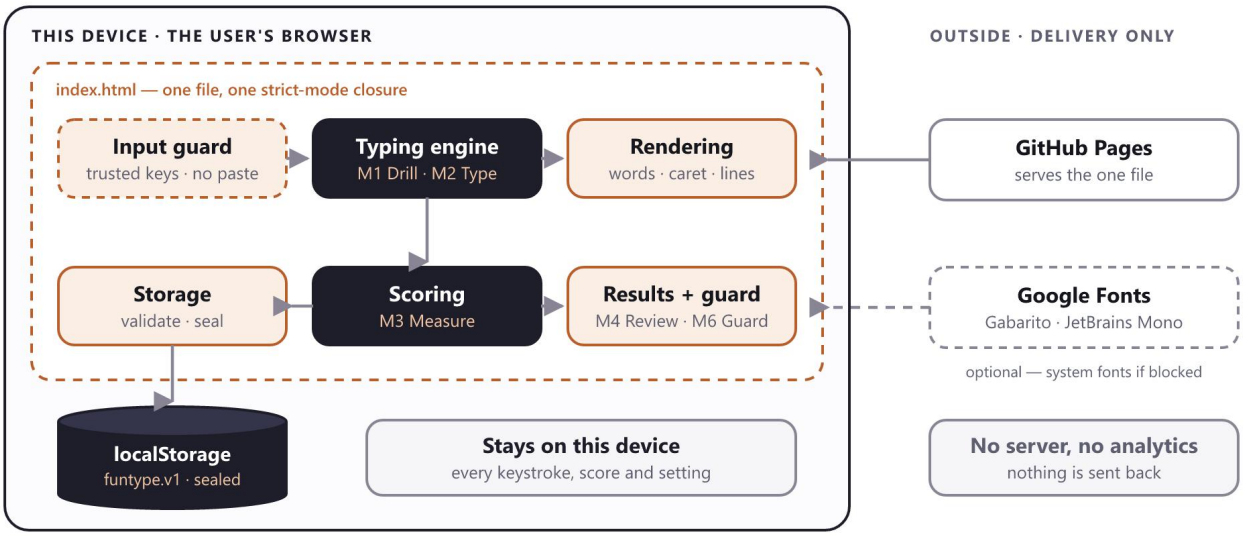


Figure 3 — System architecture — what runs where

Everything that decides a score runs inside the page, on the user's machine. The two outside services only deliver files: GitHub Pages sends the page once, and Google Fonts sends two typefaces the app can live without.

7.1 How the stack got here

One file, no build is a constraint, not an accident. It buys four things:

- The app opens from a web host, a `file://` path, a USB drive or an email attachment, identically.
- There is no toolchain to rot. A build that works today works in five years, because there is nothing to reinstall.
- The whole program can be read in one sitting, which is why the measurement model can be audited rather than trusted.
- Deployment is a file copy.

The cost is real: every new feature puts pressure on the single-file rule. M8 in Phase 4 is the first feature large enough to justify breaking it, and Risks and mitigations (§14) carries that as a decision rather than a drift.

7.2 Files

File	Size	What it is
`index.html`	45,953 B	**The app.** Markup, styles and logic in one file. The only file to edit
`funtype.html`	45,806 B	The same page without the outer HTML wrapper — the copy published as a Claude Artifact. Regenerated, never hand-edited
`logo.svg`	776 B	Full lockup: keycap mark and wordmark. Needs Gabarito to render the text
`logo-mark.svg`	530 B	The mark alone — pure geometry, safe anywhere
`logo-favicon.svg`	515 B	The mark retuned for 16–24 px; used below about 28 px
`README.md`	6.0 KB	Modes, keys, formulas, tamper guards and design notes
`docs/PROPOSAL.md`	—	This proposal, the source of the Word and PDF editions
`docs/FunType-Project-Proposal.docx`, `.pdf`	—	This proposal formatted for sharing; generated, never hand-edited
`.nojekyll`	0 B	Tells GitHub Pages to serve the files as they are

7.3 Inside the file

The script is one strict-mode closure, so nothing it defines is reachable from the page or the console.

Section	Lines	Responsibility
Styles	11–156	Eleven colour tokens, layout, the caret, the results grid, reduced motion
Markup	158–222	Header and records, settings bar, typing stage, footer, the hidden input
Content	228–285	Word list, passages, code lines, eight themes, chart colours, mode lengths
State and storage	288–371	Test state, validation, the seal, `load` and `save`
Theme	374–398	Applying a theme's tokens; building the swatch row
Word generation	401–447	Picking words; adding punctuation and numbers
Rendering	450–530	Drawing words, colouring characters, moving the caret, scrolling lines
Typing	533–572	Applying each key to the state
Clock and scoring	575–649	Timing, per-second samples and every metric
Results, chart, records	652–809	Results screen, the SVG chart, the records bar and the on-screen guard
Lifecycle and input	812–924	New tests, focus and idle blur, keyboard and input listeners, start-up

7.4 The render loop

A keystroke repaints one word and moves the caret with a CSS transform; nothing else on the page is touched. Line height is measured once and re-measured only when the window resizes or the fonts finish loading. The pace chart is inline SVG built once, at the finish — there is no charting library.

7.5 Deployment

Pushing to `main` publishes: GitHub Pages serves the repository root, and `.nojekyll` stops it processing the files. The Artifact copy is regenerated by stripping the first five and last two lines of `index.html`:

```
sed '1,5d' index.html | sed '$d' | sed '$d' > funtype.html
```

That recipe counts lines, so nothing may ever be inserted above line 6 of `index.html`. When the favicon was added on 22 September it went on line 7 for exactly this reason.

8 Data model and storage design

Everything FunType remembers is one JSON object under one `localStorage` key, `funtype.v1`, sealed with a hash of its own contents.

8.1 The saved record

```
{
  "mode": "time",
  "amount": { "time": "30", "words": "25", "text": "medium", "code": "js" },
  "punctuation": false,
  "numbers": false,
  "theme": "dusk",
  "bests": { "time-30": { "wpm": 78, "acc": 96, "at": 1758499200000 } },
  "tests": 42,
  "seal": "3kq9zv1x0b2"
}
```

Field	Type	Accepted values
`mode`	string	`time`, `words`, `text`, `code`
`amount`	object	One allowed length per mode — anything else falls back to the default
`punctuation`, `numbers`	boolean	Exactly `true`; anything else reads as `false`
`theme`	string	One of the eight theme names
`bests`	object	Keys matching the grammar below; `wpm` an integer 0–400, `acc` an integer 0–100, `at` a finite timestamp
`tests`	integer	A positive safe integer
`seal`	string	The hash of every other field — see Security and privacy (§9)

8.2 The personal-best key

A key is `mode-length`, then `-p` if punctuation was on, then `-n` if numbers were on. It must match `^(\w+)-(\w+)(-p)?(-n)?$`, and the length must be one the mode actually offers — so `time-45` or `code-rust` is dropped on load even though it fits the pattern.

8.3 The best is the only trace of a run

FunType keeps the best result per configuration and the count of tests run. It does not keep keystrokes, per-run history or samples. That keeps the save small — well under a kilobyte for a typical user — but it has one consequence that shapes the whole plan: **a stored best can never be recomputed.** If a formula in The measurement model (§3.2) changed, every best would silently mean something different.

The rule is therefore absolute. Any change to a formula bumps the key from `funtype.v1` to `funtype.v2`, and old records are retired honestly rather than misread.

8.4 Loading is defensive

- A save that cannot be parsed is ignored, and the app starts with defaults.
- A save whose seal does not match is discarded whole and replaced with a clean one.

- Every field is then checked against the values the app itself could have written. A bad field falls back to its default; it cannot stop the app starting.
- Nothing read from storage is ever placed on the page without being escaped.

8.5 Migration is a first-class concern

Two migrations have been designed so far:

1. **Pre-seal saves.** Saves written before sealing shipped carry no seal. They are accepted once — and then sealed — but only if every record in them is older than 23 September 2026, 01:00 UTC. A hand-made unsealed save with a recent record is refused.
2. **History in Phase 4.** Keeping the last twenty runs per configuration changes the shape of the save. The plan is to read `funtype.v1`, carry every best forward, write `funtype.v2`, and keep the old key untouched until the new one has been verified equal.

9 Security and privacy

A local-only trainer has no server to breach and no account to steal, so its threat model is short — but it is not empty. The realistic attacks are on the score itself: a script that types, a pasted test, an edited save, an edited screen. The work of 23 September 2026 (commit `a14402c`) closed each of them to casual attempts. A fix on 14 October 2026 (commit `8b795dc`) stopped the on-screen guard fighting browser extensions: where an extension also edited the results screen, the two undid each other in a loop that froze the last second of a test.

9.1 Threat model

Threat	How it would arrive	Status
Scripted typing	A console loop dispatching fake `KeyboardEvent`s	**Closed — only events the browser marks `isTrusted` are read**
Pasted or dropped text	Copying the visible words into the hidden input	**Closed — `paste` and `drop` are cancelled**
An impossible speed	Any trick that finishes a test absurdly fast	**Closed — over 400 WPM is shown as *not recorded* and never saved**
A hand-edited save	DevTools → Application → Local Storage	**Closed — the seal no longer matches, so the save is discarded**
Markup injected through the save	A crafted value written into `funtype.v1`	**Closed — every field validated, every output escaped**
A neighbouring app on the same origin	Another project on `kaonhew02.github.io` writing to the key	**Closed — the same validation and seal**
An edited score on screen	Inspect Element, or a line pasted into the console	**Closed — the original words and numbers are put back the moment they change**
A modified copy of the app	Downloading the public source and changing it	Out of scope — it changes only that person's own screen

9.2 Controls in place

- **Trusted input only.** Both the key handler and the hidden input's handler return at once for any event without `isTrusted`, which a script cannot set.
- **No paste, no drop.** The words are on screen, so pasting them would type the whole test at once.
- **A ceiling on the possible.** `MAX_WPM = 400` is well past any human run. A faster result is shown with a *not recorded* badge and never becomes a best.
- **A sealed save.** Every save carries `seal`, a 53-bit hash (the cyrb53 construction) of all its other fields with a fixed prefix. On load the hash is recomputed; any difference discards the whole save.
- **Validation and escaping.** Each stored field is checked against what the app could have written, and every value shown on the page passes through an HTML escape.
- **A guarded screen.** The records bar and the results screen are drawn through one function that remembers what it drew, and a `MutationObserver` puts back the original the instant anyone changes the words or numbers in it. It compares text only, so dark-mode and other styling extensions are left alone, and it stands down for a node that is rewritten more than three times in a second — a translator, not a person — rather than fight it in a loop. The result buttons use a delegated click handler, so they keep working after the guard rebuilds them.

- **A sealed script.** The whole application is one closure; no function or piece of state is reachable from the console by name.

9.3 Privacy by construction

FunType has no backend, no analytics, no error reporting, no cookies and no identifiers. The only third-party request is the Google Fonts stylesheet, which — like any font service — sees the visitor's IP address and nothing else. Typing, scores and settings never leave the browser.

On GitHub Pages every project under one account shares a single origin, `kaonhew02.github.io`, and so a single `localStorage`. FunType therefore treats its own save as untrusted input, which is why the validation above exists even though no attacker is expected.

9.4 What is not protected, and why

- **The source is public.** Anyone who reads it can compute a valid seal or run a changed copy. The guards are deterrents, not locks — and since nothing is ever sent anywhere, a cheat can only fool its author.
- **A trusted shared leaderboard is not possible here.** It would need a server that replays and checks each run. That is out of scope, and the reason leaderboards are.
- **A shared computer shows its records** to anyone using the same browser profile. Accepted: they are typing scores, and the operating-system account is the lock.

10 Brand, UI and UX design

The interface is one calm screen: the words are the largest thing on it, and everything else fades while you type.

10.1 Identity

Name	FunType
Mark	A keycap whose legend is a text caret — the same caret the user chases across the screen
Wordmark	Gabarito ExtraBold, tracking −0.022 em; "Fun" in the accent, "Type" in the text colour
Tagline	*train the hands, the words follow.*
Voice	Lower-case, plain labels: *next test*, *repeat this text*, *restart test*

10.2 The mark

- The caret blinks on the same 1.05 s beat as the typing caret, and the cap face sinks 2.6 px on hover or press — because the mark is also the restart button.
- **Redrawn on 22 September 2026.** The first version set the caret in a dark inset dish, where it filled about 10% of the frame and blurred into a smudge at the 38 px it ships at. The redraw removed the dish and grew the caret to 63% of the cap height: four shapes down to two.
- `logo-favicon.svg` pushes the cap to the edges and thickens the caret for 16–24 px. The favicon is embedded in the page as a `data:` URI, so `index.html` still works with nothing beside it.

10.3 Palette

Every colour in the app is one of eleven CSS custom properties; no component carries a hard-coded colour. A theme is therefore a small data object, and each one supplies its own `accent-ink` so the mark recolours correctly in all eight.

Theme	Kind	Background	Accent
dusk (default)	Dark	`#16161e`	`#ff9e64` apricot
daylight	Light	`#eceef4`	`#cf6a2c`
matcha	Dark	`#1b2420`	`#9ccf5e`
cobalt	Dark	`#0f1729`	`#5ec8ff`
nordfall	Dark	`#2e3440`	`#88c0d0`
bubblegum	Light	`#fdf0f3`	`#e5527e`
mono	Dark	`#0e0e10`	`#f5f5f5`
terminal	Dark	`#050a05`	`#35ff4d`

The chart colours are fixed across themes — net `#d8792e`, raw `#6189e0`, errors `#cf4a66` — checked for colour-blind separation on both dark and light surfaces. A legend that changed meaning between themes would be a chart nobody could read twice.

10.4 Typography

Use	Typeface	Fallback	Why
Interface	Gabarito, 400–800	Segoe UI, system-ui	A warm geometric sans that keeps the chrome from feeling clinical
Everything typed, all numbers	JetBrains Mono, 300–700	Cascadia Mono, ui-monospace	Equal character widths, so the caret moves evenly and error marks line up

Scores use tabular figures, so digits do not jump sideways as they change.

10.5 Accessibility

- Every control is a real button with a 2 px accent focus ring. Because `tab` and `space` are test keys, the controls are reached with `shift` + `tab` and pressed with `enter` — a gap listed under known issues in Risks and mitigations (§14).
- Toggles, modes and theme swatches report their state with `aria-pressed`; swatches are labelled by name; the typing input and the settings bar are labelled.
- The pace chart is an image with a text label giving the final net and raw figures.
- The live counter is deliberately silent to screen readers, so they are not interrupted every second.
- All animation and transitions switch off under the system's reduced-motion setting.

10.6 Layout

Area	Contents	Behaviour
Header	Mark and wordmark; three record chips	The mark restarts; the chips update after every test
Settings bar	Toggles · modes · lengths	Fades out while a test runs
Stage	Live counter, three lines of words, *restart test*	Replaced by the results when a test ends
Footer	Key hints, eight theme swatches, the tagline	Key hints hide on narrow screens

The page is capped at 1,060 px wide. At 680 px and below the results stack into one column, the record chips take the full width, and the typing text scales down with the viewport.

11 Technology stack and tooling

Layer	Choice	Why
Structure	HTML5	Semantic regions; no templating needed for one screen
Styling	CSS custom properties, flexbox and grid	Themes as data; no CSS framework
Logic	Vanilla JavaScript, strict mode, one closure	Readable in one sitting; nothing reachable from the console
Charts	Inline SVG generated at the finish	One chart does not justify a library
Storage	Web Storage (`localStorage`)	One small JSON object; synchronous and universal
Integrity	`Event.isTrusted`, `MutationObserver`, cyrb53 hash	Built into the browser, or twenty lines of code
Fonts	Google Fonts — Gabarito, JetBrains Mono	The only external asset, with system fallbacks
Hosting	GitHub Pages	Free for a public repository; serves static files
Version control	Git, github.com/KaonHew02/FunType	Public source, which the security model assumes

11.1 The one external asset, and why it is acceptable

Google Fonts supplies the two typefaces, preconnected so they arrive with the page. If the request fails — offline, blocked, or opened from a USB drive — the styles fall back to Segoe UI and Cascadia Mono and the app stays fully usable. Only the brand suffers, never the function.

11.2 What was deliberately not used

- **No framework.** One screen with a handful of states does not need a virtual DOM, and a framework would end the single-file rule on day one.
- **No bundler or package manager.** There is nothing to bundle, and a `node_modules` folder is the first thing to rot.
- **No charting library.** The chart is about fifty lines of code producing SVG.
- **No service worker.** Offline install is on the roadmap, but a worker is a second file and brings its own stale-cache trap.

11.3 Development workflow

1. Edit `index.html` directly — there is no build.
2. Open the file in a browser by double-clicking it; no local server is needed.
3. Walk the release checklist in Testing and quality assurance (§13).
4. Regenerate `funtype.html` with the `sed` recipe, and update `README.md` if behaviour changed.
5. Commit and push to `main`; GitHub Pages republishes the root.

12 Project plan and timeline

Three phases are complete. Phase 4 is what this proposal asks approval for: ten weeks from 28 September to 6 December 2026, closing the open objectives and adding coaching.

12.1 Phases 1–3 — delivered

Phase	Dates	Delivered
1 · Build	11 Sep 2026	The complete trainer in one file: four modes, the measurement model, the results chart, per-configuration bests, eight themes and the keycap mark (commit `2432633`)
2 · Brand	22 Sep 2026	The mark redrawn so the caret survives small sizes, a dedicated favicon (`d812abf`), and the first project proposal (`6465d3b`)
3 · Hardening	23 Sep 2026	Trusted-input check, paste block, 400 WPM cap, sealed save and on-screen guard (`a14402c`); the proposal rewritten (`c742315`)

Five commits across those three phases, and the app is live on GitHub Pages. The build was small because the scope was held small on purpose: every feature that was not a drill, a number or a record was left out. One fix has shipped since: on 14 October 2026 the on-screen guard stopped fighting browser extensions, which had frozen the last second of a test (`8b795dc`).

12.2 Phase 4 — proposed

Week	Dates	Work	Deliverable
1	28 Sep – 4 Oct	**Freeze the model.** Decide open questions 1 and 2 in Risks and mitigations (§14); fix the known issues; build the scoring harness	Measurement model frozen; objective O13 met
2–3	5–18 Oct	**Content depth.** Word list to at least 1,000 unique words; passages from 6 to 15; code from 9 to 18 lines, adding TypeScript, Go and shell	Objective O14 met
4–5	19 Oct – 1 Nov	**History and export.** Last 20 runs per configuration with a trend line; export and import of the save; `funtype.v1` → `funtype.v2` migration	Every v1 best readable after migration
6–9	2–29 Nov	**M8 Coach.** Per-key error and delay profile across recent runs; a drill weighted to the weakest keys; a coaching panel on the results	Weak keys named and drilled
10	30 Nov – 6 Dec	**Release.** Browser regression, the eight-theme visual pass, accessibility re-check, README and proposal refresh	v1.0 live; objective O12 met

12.3 Milestones

ID	Milestone	Due	Gate
MS-1	Measurement model frozen	2026-10-02	Open questions 1 and 2 decided in writing
MS-2	Scoring harness green	2026-10-04	Every metric asserted against a known keystroke sequence
MS-3	Content pool tripled	2026-10-18	1,000+ unique words; 15 passages; 18 code lines
MS-4	History and export live	2026-11-01	A v1 save migrates with no best lost; export → wipe → import is identical
MS-5	Coaching live	2026-11-29	Weak keys named on the results and drilled on demand
MS-6	v1.0 released	2026-12-06	Live, documented and tagged

MS-1 is a hard gate. Every later milestone produces records, and records made before the model is frozen are records that may have to be thrown away.

13 Testing and quality assurance

Today FunType is tested by hand, in the browser. That has been enough to ship, but it is not enough to freeze a measurement model, so Phase 4 adds an automated harness for the one part of the code users trust most: the scoring.

Layer	Tool	Catches
Behaviour	A real browser, driven by hand through the test cases below	Broken keys, wrong colouring, a mode that never ends
Scoring	Node and jsdom, replaying fixed keystroke sequences — **proposed**	A formula that drifts; a count off by one; a space counted twice
Security	The browser console and DevTools	A scripted key that types; an edited save that survives; a score that stays edited
Layout and colour	The browser at desktop, tablet and 375 px widths, in all eight themes	Overflow, unreadable text, a mark that does not recolour

13.1 Test cases

ID	Test	Expected result
TC-01	Type a 10-word test with no mistakes	Accuracy 100%; characters `n/0/0/0`
TC-02	Leave one wrong character uncorrected	Wrong = 1; the word underlined; net below raw
TC-03	Type a wrong character, then fix it with backspace	The wrong keystroke counts once; the backspace costs nothing
TC-04	Press space two letters early	Missed = 2
TC-05	Keep typing past the end of a word	Extra stops at 12
TC-06	Backspace at the start of a word, after a correct and after a wrong word	Refused after the correct word; allowed after the wrong one
TC-07	Run `time 15`	Ends at 15 s; the words never run out
TC-08	Beat a stored best, then try a new configuration	`new best · +N`, then `first record`
TC-09	Dispatch a `KeyboardEvent` from the console	Ignored
TC-10	Paste or drop text into the test	Nothing is typed
TC-11	Edit a WPM in DevTools → Local Storage, then reload	The save is discarded; records are empty
TC-12	Change the WPM with Inspect Element	The value snaps back
TC-13	Open in a private window with storage blocked	The app works and forgets afterwards
TC-14	Block `fonts.googleapis.com`	Fallback fonts; fully usable
TC-15	Switch through all eight themes	Text legible; the mark recolours correctly
TC-16	Open `index.html` straight from disk	Behaves exactly as on the web
TC-17	Turn on reduced motion in the operating system	No caret glide, blink or fades
TC-18	Narrow the window to 375 px	Results stack; key hints hide; no sideways scroll
TC-19	Finish a test with a dark-mode or translation extension switched on	The results appear at once; the screen never freezes

13.2 The proposed scoring harness

The harness loads `index.html` into jsdom, freezes the clock, and replays fixed keystroke sequences — a clean run, a run with one wrong letter, one with an early space, one with a correction — asserting the exact net WPM, raw WPM, accuracy, consistency and character breakdown of each. There are no mocks of the scoring code, because a mock of the arithmetic is not a test of the arithmetic.

13.3 Browser matrix

Browser	Status
Chrome and Edge on Windows 11	**Primary target, used throughout development**
Firefox	**Needs re-verification each release**
Safari on macOS	**Needs verification — no Mac in the current toolset**
Any private window	**Degrades gracefully: runs normally, remembers nothing**

13.4 Two traps worth writing down

The guard also stops the tests. A key event dispatched by jsdom has `isTrusted` set to false, exactly like a console auto-typer, so the app ignores it. The harness cannot drive the page through its keyboard. It has to reach the typing functions directly — by appending a small test-only hook inside the closure before evaluating it — and that hook must never reach the live file.

The Artifact copy is cut by line count. Anything inserted above line 6 of `index.html` silently changes what the `sed` recipe strips from `funtype.html`, and the damage is not obvious. A marker-based strip is on the roadmap; until then, the head of the file stays exactly five lines long.

13.5 Release checklist

- ☐ The scoring harness is green (from Phase 4)
- ☐ TC-01 to TC-19 pass in Chrome or Edge
- ☐ A scripted key, a paste and an edited save are all refused
- ☐ All eight themes checked; the mark correct in each
- ☐ No sideways scroll at 375 px, 768 px and desktop widths
- ☐ `funtype.html` regenerated and identical to `index.html` minus its wrapper
- ☐ README updated if a key, formula or mode changed
- ☐ Storage key bumped if any formula changed
- ☐ Pushed to `main`, and the live site checked with a hard reload

14 Risks and mitigations

The architecture trades a server for privacy, simplicity and cost, and most of the risks below are the bill for that trade. They are stated plainly rather than minimised.

ID	Risk	Likelihood	Impact	Mitigation	Residual
R1	**A formula changes after users hold records**, and every stored best silently means something else	Low	Critical	Freeze the model at MS-1; any later change bumps the key to `funtype.v2`	Low
R2	**Records live in one browser only.** Clearing site data erases every best	High over a year	Medium	Export and import of the save in Phase 4	Low once export exists
R3	**The content is small enough to memorise**, so scores rise without skill rising	High	High	Content tripled in weeks 2–3	Low
R4	**The scoring has no automated tests**	Certain today	High	The scoring harness in week 1, before any other change	Low
R5	M8 Coach is large enough to break the single-file rule	Medium	Medium	Decide it explicitly at the start of week 6, not by drift	Medium
R6	Safari deletes script-written storage after 7 days without a visit	Medium	Medium	Export in Phase 4; state it plainly for Safari users	Medium — not fixable in the page
R7	The `funtype.html` strip depends on line counts	Medium	Medium	A marker-based strip	Low
R8	Another project on `kaonhew02.github.io` writes to `funtype.v1`	Low	Medium	Seal and field validation (delivered)	Low
R9	A determined person fakes a score	Medium	Low	Accepted — it fools only their own screen	Low
R10	Google Fonts is unreachable	Low	Low	System fallbacks (delivered)	Low
R11	A single developer; no second person knows the code	Medium	High	README, this proposal, and a file small enough to read in one sitting	Medium
R12	Scope creeps toward accounts or a leaderboard	Low	High	The non-objectives in Objectives and success criteria (§4)	Low
R13	**Browser extensions edit the guarded screen** — dark mode, translators — and a guard that fights them freezes the page	High	High	The guard compares text only and stands down after three rewrites in a second (delivered 14 Oct 2026)	Low

14.1 The three that deserve a decision, not just a mitigation

R1 is the only expensive one to reverse. It is why MS-1 is a hard gate, and why two open questions must be answered before week 2:

1. Should consistency sit beside accuracy in the headline, since it is the most diagnostic figure on the screen?
2. Should a personal best require a minimum accuracy — for example 90% — so a fast, careless run cannot hold the record?

R2 and R6 are the same risk. Both end with a browser that no longer holds the records. The only real answer is a copy that is not in the browser, which is why export ships in Phase 4.

R5 is a trade, not a bug. Breaking the single-file rule for coaching may be the right call. It should be made on purpose, in writing, at the start of week 6.

14.2 Known issues in the current build

Issue	Effect	Fix
The toggles still split the record key in `text` and `code` modes, though they do not change that content	`code-js` and `code-js-p` are separate records for identical text	Ignore the toggles in the key for those modes, or hide them there — week 1
The word list holds 10 duplicate entries — `door` three times; `high`, `small`, `late`, `water`, `build`, `start`, `story` and `watch` twice	Frequency slightly skewed	De-duplicated in weeks 2–3
The chart note asks for 15 seconds, but the chart appears after 2	A misleading message	Align the wording — week 1
`consistency()` opens with a filter that keeps every element	Dead code; the result is correct	Remove it — week 1
`tab` and `space` are test keys, so the settings are reached with `shift` + `tab` and pressed with `enter`	Hard to discover for keyboard-only users	A visible hint, checked in the week 10 accessibility pass

15 Resources and budget

FunType costs RM 0 a month to run today, and Phase 4 keeps it there. The only real resource is developer time.

15.1 People

Role	Who	Commitment
Developer, designer and tester	One person	About 5 hours a week for 10 weeks — **about 50 hours**
Reviewer and product owner	The same person, practising on the app	Continuous — FunType is used while it is built
Volunteer typists	Three to five people	One session each at MS-1 and MS-5, reading the formulas and saying whether the scores feel fair

A second developer is not proposed. The codebase is one readable file, and Phase 4 is not large enough to split.

15.2 Tools and infrastructure

Item	Cost	Note
GitHub repository and Pages hosting	RM 0	Free for public repositories
Storage	RM 0	The user's own browser
Google Fonts	RM 0	Open-licence typefaces
Browsers and DevTools	RM 0	Chrome, Edge and Firefox
Node.js and jsdom for the harness	RM 0	Development only; nothing ships
Claude Code	Existing subscription	AI pair-programming; co-author on every commit so far
Current total	**RM 0 / month**	

15.3 Optional and conditional costs

Item	Cost	When it applies
Custom domain	About RM 60 / year	Only if the `github.io` address is not wanted
A server to verify a shared leaderboard	About US$5–7 / month (≈ RM 21–30)	Not proposed — listed to show what a leaderboard would really cost

Ringgit figures are approximate, converted at about RM 4.25 to the US dollar; they are for sizing, not budgeting to the sen.

15.4 Three-year total cost of ownership

Scenario	Year 1	Years 2–3	3-year total
Stay as proposed	RM 0	RM 0	**RM 0**
With a custom domain	RM 60	RM 120	**RM 180**
With a verification server (not proposed)	About RM 306	About RM 612	**About RM 918**

15.5 What the budget does not buy

No paid analytics, no error-reporting service, no CI runner, no design tooling and no font licence. Everything in the stack is free at this scale because the app is deliberately small enough to stay there.

16 Future roadmap

After v1.0 the work splits into two groups: things worth building, and things deliberately not built.

16.1 Candidates for v1.1 and beyond

Item	Value	Cost
Shareable result links	A finished run encoded in the URL fragment, linkable with no server	Low — but it edges toward a leaderboard feel
A "hard words" toggle	Drills from a low-frequency list for typists who have outgrown the common one	Low — pure data
Custom practice text	Type a passage of the user's own choosing	Low — entered in a settings box, not pasted into the test
Install as a web app	Offline start and a home-screen icon	Medium — a service worker ends the single-file rule
A marker-based Artifact strip	Removes the line-count trap in R7	Low — a one-line script
Per-finger view	Weak keys grouped by the finger that owns them, on top of M8	Medium

16.2 Deliberately not built, and why

These have each been considered and declined. The reasons are recorded so they are not re-proposed as if new:

- **A global leaderboard.** It is only as honest as the server that checks each run, and FunType has none.
- **Charging for backspace.** It teaches people to leave mistakes in, which is the habit the product exists to break.
- **Gross WPM as the headline.** It rises as accuracy falls.
- **Accounts and sync.** They cost the privacy claim, and export covers the real need — records surviving a new laptop.
- **Lessons and a curriculum.** A different product for a different user.

17 Conclusion and approval

FunType already works. It is live, it measures honestly with every formula published, it keeps a separate record for every configuration, it costs RM 0 a month, and since 23 September it resists the ways a local-only trainer's scores can actually be faked. What this proposal asks for is not a build from nothing — it is ten weeks to prove the numbers by test, make the content deep enough that the numbers stay honest, and then turn them into coaching.

The case rests on three things:

1. **The foundation is sound and paid for.** The measurement model is defined, implemented and public; bests are kept per configuration; the save is validated and sealed. That is the hard part, and it is done.
2. **The remaining scope is small and known.** One week to freeze and test the model, two for content, two for history and export, four for coaching, one to release.
3. **The downside is bounded.** If M8 proves harder than planned, Phase 4 still delivers a tested measurement model, three times the content, history and export — at RM 0.

17.1 The decision requested

Approve	Phase 4 as scoped — 10 weeks, about 50 hours, one developer
Budget	RM 0 committed
Decide by	2026-09-25
Start	2026-09-28

If Phase 4 is not approved, the recommendation is to do week 1 anyway: answer the two open questions, fix the known issues and build the scoring harness. It is about five hours, and it protects every record that users make from then on.

17.2 Sign-off

Role	Name	Decision	Date
Product owner			
Developer			
Reviewer			

17.3 Sources

Every figure in this proposal is taken from the FunType repository as it stands on 14 October 2026 (commit `8b795dc`, which is also where the application code last changed) — `index.html`, `README.md`, the three logo files and six commits of git history from 11 September to 14 October 2026 — and from the live site at

kaonhew02.github.io/FunType, checked the same day. No external source was consulted. The comparisons in Background and problem statement (§2) and the costs in Resources and budget (§15) are stated from general knowledge and should be checked before the document is shown outside the project.

18 Glossary

Term	Meaning
WPM	Words per minute, where one "word" is five characters
Net WPM	Correct characters ÷ 5 ÷ minutes — FunType's headline figure
Raw WPM	Every character typed, right or wrong, ÷ 5 ÷ minutes
Gross WPM	Raw WPM used as a headline; rises as accuracy falls
Accuracy	Correct keystrokes as a share of all keystrokes; backspaces are not counted
Consistency	100 × (1 − standard deviation ÷ mean) of the per-second pace
Extra	A character typed past the end of a word
Missed	A character left untyped because space was pressed early
Configuration key	The name a personal best is stored under, such as `time-30-p`
`localStorage`	The browser's small key-value store for a site; where FunType keeps its save
Origin	The scheme, host and port a page is served from; pages on one origin share storage
Seal	A hash of the save's own contents, stored beside it, so an edit is detectable
cyrb53	A small, fast, non-cryptographic 53-bit string hash
`isTrusted`	A flag the browser sets on events caused by the user, which scripts cannot set
`MutationObserver`	A browser feature that reports changes to part of the page
IME	Input method editor — software that composes characters, as on phones and for many languages
Reduced motion	An operating-system setting asking apps to minimise animation
GitHub Pages	GitHub's free static hosting, serving FunType at `kaonhew02.github.io/FunType`
Claude Artifact	A page published on claude.ai; `funtype.html` is the copy made for it